Feature Engineering Report

Employee Attrition Dataset — HR Analytics Feature Engineering Pipeline

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Field	Detail
Dataset	5,000 employee records, 39 original variables
Target Variable	Attrition (0 = Stayed, 1 = Left)
Subject	Feature Engineering — Graded Assignment
Deliverables	Jupyter Notebook (.ipynb) + this Feature Engineering Report (PDF)

1. Executive Summary

This report summarizes the feature engineering pipeline developed for a synthetic but realistic HR attrition dataset containing **5,000 employee records** and **39 variables**. The objective was not to build a predictive model, but to transform raw, imperfect HR data — with missing values, skewed distributions, outliers, and redundant/correlated features — into a clean, well-justified, machine-learning-ready dataset supported by a reusable preprocessing pipeline.

The pipeline addressed four missing-value columns, capped outliers in compensation and tenure fields, log-transformed skewed monetary/count features, applied type-appropriate categorical encoding, constructed eight new business-driven features, removed statistically and algebraically redundant columns, evaluated (and consciously declined) PCA-based dimensionality reduction, and finally packaged every transformation into a single reusable `scikit-learn` pipeline.

Class Balance

The target variable is imbalanced: **80.9% of employees stayed** and **19.1% left**. This is noted for downstream modelling awareness (e.g. the need for class-weighting or resampling at training time) but is not addressed inside this feature engineering pipeline, which is scoped to data preparation rather than model training.

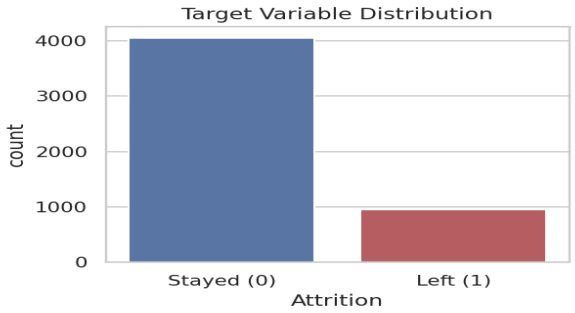


Figure 1. Attrition class distribution (~19% attrition).

2. Key Findings from Exploratory Analysis

2.1 Missing Values

Four columns had missing values in the 5–7% range: Training_Hours, Annual_Bonus, Manager_Rating, Environment_Satisfaction. Attrition rates were checked between missing and non-missing subgroups for each column and found to be similar, indicating a **Missing-At-Random (MAR)** pattern rather than informative missingness (e.g. employees withholding data related to their attrition intent).

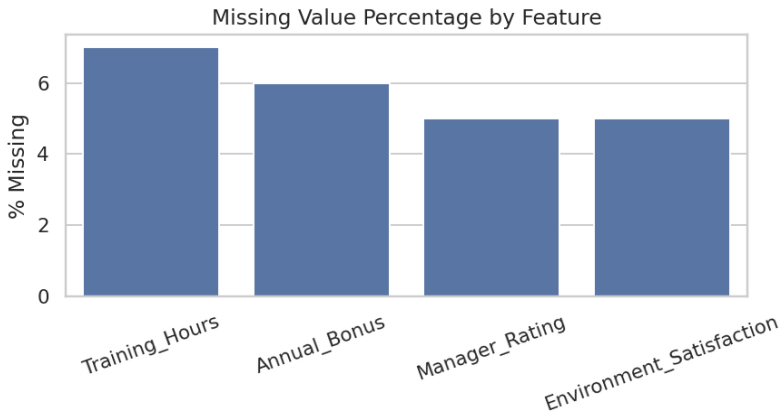


Figure 2. Missing value percentage by feature.

2.2 Skewed Distributions

Compensation and tenure-related features showed strong right skew — a common characteristic of monetary and count data, where most employees cluster at lower values with a long tail of senior or highly-compensated employees.

Feature	Skewness
Monthly_Salary	2.14
Annual_Bonus	3.06
Stock_Option_Value	3.38
Annual_Compensation	2.16
Distance_From_Home	1.90
Overtime_Hours	1.61
Years_in_Current_Role	1.92
Years_at_Company	1.15
Certification_Count	1.39
Monthly_Expenses	1.60

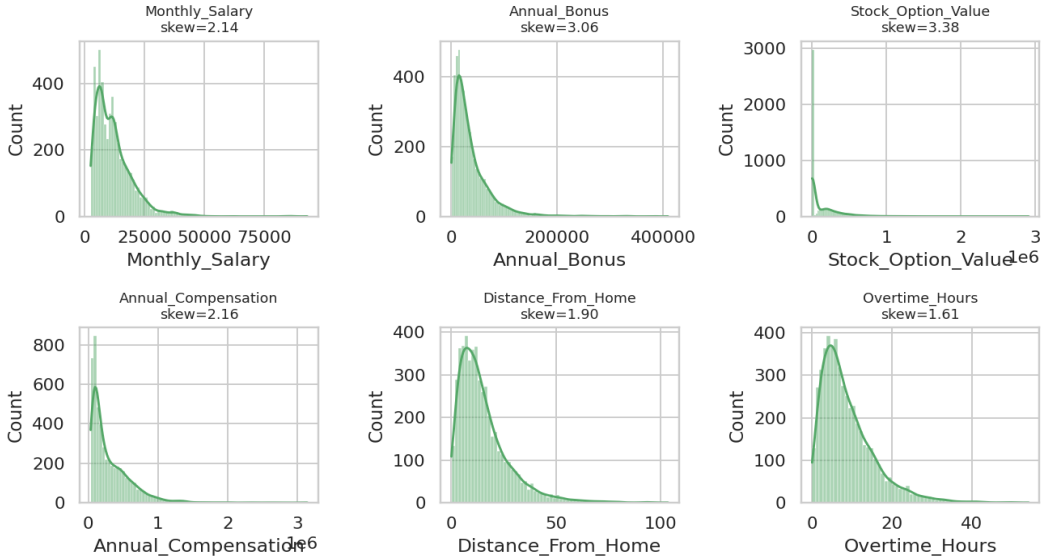


Figure 3. Distribution of key right-skewed features.

2.3 Outliers

Outliers were assessed using the IQR method (1.5×IQR beyond Q1/Q3). Rates were moderate (1–6.5%) and concentrated in compensation and tenure fields — plausible extreme values from senior or high-earning employees rather than data errors.

Feature	Outlier Count	Outlier %
Stock_Option_Value	316	6.32%
Certification_Count	298	5.96%
Annual_Bonus	261	5.22%
Years_in_Current_Role	244	4.88%
Distance_From_Home	224	4.48%
Overtime_Hours	195	3.90%
Monthly_Expenses	188	3.76%
Monthly_Salary	163	3.26%
Annual_Compensation	156	3.12%
Years_at_Company	120	2.40%

2.4 Correlation & Redundancy Signals

A correlation heatmap across all numeric features surfaced several tightly related clusters — most notably compensation components, tenure/promotion measures, and employee survey/engagement scores — which directly informed the redundancy-removal decisions in Section 4.

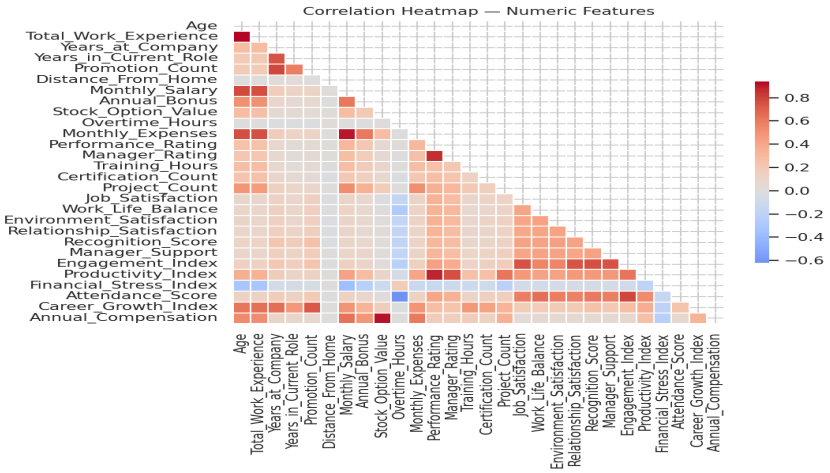


Figure 4. Correlation heatmap of numeric features (lower triangle).

Feature 1	Feature 2	Correlation
Age	Total_Work_Experience	0.941
Stock_Option_Value	Annual_Compensation	0.917
Monthly_Salary	Monthly_Expenses	0.909
Performance_Rating	Productivity_Index	0.882
Performance_Rating	Manager_Rating	0.863
Years_at_Company	Promotion_Count	0.793
Engagement_Index	Attendance_Score	0.785
Age	Monthly_Salary	0.777
Age	Monthly_Expenses	0.770
Total_Work_Experience	Monthly_Salary	0.765

Notably, **Annual_Compensation** was verified to be an *exact* algebraic sum of **Monthly_Salary × 12 + Annual_Bonus + Stock_Option_Value** (residual ≈ 0), confirming it as a fully derived, redundant column rather than an independently-collected measurement.

3. Feature Engineering Approach & Justification

3.1 Missing Value Treatment

Median imputation was applied to all four columns with missing values. Median was preferred over mean because these columns are either skewed (Training_Hours, Annual_Bonus) or ordinal integer scales (Manager_Rating, Environment_Satisfaction), where a mean could produce a non-representative, non-integer value. A binary "was-missing" indicator flag was added for each column as a low-cost, low-risk feature in case any subtler missingness pattern exists beyond the aggregate check.

Alternative considered: KNN/regression-based imputation could exploit correlations with other features for more accurate estimates, but was judged unnecessary given the modest missing rate (≤7%) and the confirmed MAR pattern — simple, reproducible median imputation was preferred for pipeline simplicity.

3.2 Outlier Treatment

Outliers were **capped (winsorized) at the 1st and 99th percentiles** rather than removed. This limits the leverage of extreme values on scale-sensitive models while preserving every row — important given the dataset's already-imbalanced target. Percentile-based capping was chosen over the more aggressive IQR-based bounds, which would have trimmed values too close to the bulk of highly-skewed distributions like Stock_Option_Value.

3.3 Transformation of Skewed Features

Features with |skew| > 1 after capping were **log1p-transformed** to reduce skew and produce more symmetric distributions, benefiting linear and distance-based models. Tree-based models are unaffected by this monotonic transform, so it is a safe default across model families.

3.4 Categorical Encoding

Encoding strategy was matched to variable semantics:

- **Ordinal categoricals** (Education_Level, Compensation_Band, Experience_Band) were mapped to an explicit, manually defined integer order (e.g., High School < Bachelor < Master < Doctorate), preserving their natural rank — which one-hot or arbitrary label encoding would have discarded.
- **Nominal categoricals** (Gender, Marital_Status, Department, Job_Role, Business_Travel, Work_Mode) were one-hot encoded, since they have no inherent order and one-hot avoids imposing a false rank.

Alternative considered: Target/mean encoding was considered for the nominal columns but rejected given their low cardinality (3–7 categories) — one-hot encoding is simpler and avoids any target-leakage risk that mean-encoding schemes must otherwise guard against with careful cross-validation.

3.5 Scaling

`StandardScaler` (zero mean, unit variance) was selected as the default scaler for the final pipeline, applied after capping and log-transformation. It is the standard, safe choice for linear models, PCA, and distance-based methods. Scaling was deliberately deferred to the pipeline stage (fit only on training data) to avoid data leakage in a real train/test workflow, rather than being applied destructively during exploratory transformation.

4. Feature Construction & Feature Selection

4.1 Engineered Features

Eight new features were constructed, each tied to a specific HR business hypothesis about attrition drivers:

Feature	Formula	Business Rationale
Tenure_Ratio	Years_at_Company / (Total_Work_Experience+1)	Low ratio for an experienced hire suggests they aren't yet anchored at this company.
Promotion_Rate	Promotion_Count / (Years_at_Company+1)	Slow promotion velocity is a classic driver of career-stagnation attrition.
Role_Stability	Years_in_Current_Role / (Years_at_Company+1)	Distinct stagnation proxy — time stuck in the same role, not just the same company.
Income_per_Experience	Monthly_Salary / (Total_Work_Experience+1)	Flags experienced employees who may feel under-compensated relative to peers.
Overtime_Intensity	Overtime_Hours / (Project_Count+1)	High overtime given low workload signals inefficiency/burnout risk, not just busyness.
Wellbeing_Composite	Mean of 4 correlated survey scores	Consolidates redundant survey items into one robust wellbeing signal.
Age_at_Joining	Age − Years_at_Company	Separates career stage at hire from current age.
High_Commute_Flag	Distance_From_Home > 75th percentile	Commute burden is plausibly a threshold effect, not purely linear.

4.2 Redundancy Removal

Correlation analysis (Section 2.4) and a manually-computed Variance Inflation Factor (VIF) check — implemented via `sklearn.linear_model.LinearRegression` regressions since `statsmodels` was unavailable in this environment — confirmed several redundant clusters. Notably, one cluster of survey/engagement items produced **infinite VIF**, meaning those columns are near-exactly reconstructable from one another.

Feature(s)	Redundancy Evidence	Decision
Annual_Compensation	Exact derived sum of 3 other columns (residual ≈ 0)	Dropped
Age	r = 0.94 with Total_Work_Experience	Dropped (Age_at_Joining recovers demographic signal)
Job_Satisfaction, Work_Life_Balance, Relationship_Satisfaction, Recognition_Score	Infinite VIF with Engagement_Index / Manager_Support	Dropped (folded into Wellbeing_Composite)
Performance_Rating, Manager_Rating	r = 0.76-0.88 with Productivity_Index	Dropped (coarse duplicates of a continuous index)

Additionally, once raw skewed columns had been log-transformed (Section 3.3) and the ordinal columns ordinally encoded (Section 3.4), the original raw/string versions were dropped to avoid keeping two representations of identical information.

4.3 Feature Relevance & Final Selection

Relevance to Attrition was assessed using two complementary, model-agnostic-ish heuristics: **Mutual Information** (captures non-linear relevance) and **Random Forest feature importance** (a standard relevance ranking across mixed feature types). These were used strictly as a *selection heuristic*, not as a claim about predictive accuracy, consistent with the assignment's evaluation scope.

A feature was only removed for low relevance if **both** methods agreed it carried near-zero signal — this conservative rule avoids discarding a feature that one method happens to under-rate. Several engineered features from Section 4.1 (e.g. Wellbeing_Composite, Overtime_Intensity, Promotion_Rate) ranked among the more relevant features in this check, supporting their inclusion.

Final selected feature set: 54 predictor columns (plus the Attrition target), spanning transformed numeric features, ordinal-encoded categoricals, one-hot encoded nominal categoricals, missingness indicator flags, and the eight engineered features.

5. Dimensionality Reduction Analysis

With ~54 predictor columns for 5,000 rows (a healthy ~90:1 row-to-feature ratio), and most severe multicollinearity already resolved via explicit redundancy removal (Section 4.2), dimensionality reduction was not assumed to be necessary — but was tested empirically via PCA rather than skipped outright.

PCA on the standardized final feature set required **31 of 54 components** to reach 90% cumulative explained variance (36 for 95%) — variance was spread fairly evenly across many components rather than concentrating in a small number, which is the expected result once linear redundancy has already been removed by hand.

Decision: PCA was evaluated but not adopted for the final feature set

- **Interpretability:** HR stakeholders need to trace attrition drivers back to concrete, explainable features (e.g. “overtime intensity”) — PCA components are linear blends of many original features and lose that direct interpretability.
- **Evenly spread variance:** needing 31/54 components for 90% variance means there is no small set of dominant components PCA could cleanly compress without real information loss.
- **One-hot columns don't compress well** under linear PCA, since binary indicators for Department/Job_Role contribute close to orthogonal variance and can distort component loadings.
- **Row-to-feature ratio is already healthy** (~90:1), so the curse-of-dimensionality risk PCA typically mitigates isn't a pressing concern here.

For completeness, a narrower, scoped application of PCA restricted only to the continuous numeric block (excluding one-hot columns) is documented as a viable alternative if a future modelling step specifically requires a compact numeric representation (e.g. for distance-based clustering).

6. Reusable Preprocessing Pipeline

Every transformation decision above was packaged into a single reusable flow: **raw employee data** → **engineer_features()** → **preprocessing_pipeline (ColumnTransformer)**.

- **Skewed numeric columns**: median impute → percentile-cap → log1p → StandardScaler.
- **Other numeric/ordinal-scale columns**: median impute → StandardScaler.
- **Ordinal categoricals**: explicit-order OrdinalEncoder → StandardScaler.
- **Nominal categoricals**: OneHotEncoder (drop-first, unknown categories ignored).
- **engineer_features()**: a standalone function recreating the 8 engineered ratio/composite features, applied before the ColumnTransformer.

The pipeline was validated by fitting on the full dataset and successfully transforming a fresh 20-row sample batch without errors, confirming it is reusable and leakage-safe (all statistics are learned via `.fit()` and reapplied via `.transform()` rather than recomputed on new data).

7. Summary & Reflection

Challenges Encountered

- Verifying *exact* algebraic redundancy (Annual_Compensation) required going beyond the correlation matrix to confirm the derivation numerically, rather than relying on correlation strength alone.
- Ordering feature construction before redundancy removal was necessary, since several engineered ratios (e.g. Income_per_Experience) depend on raw columns that were later dropped or transformed.
- Distinguishing genuinely “low relevance” features from single-metric false negatives required requiring agreement between two different relevance measures before dropping anything.
- statsmodels was unavailable in this environment; VIF was reimplemented manually via sklearn linear regressions, reproducing the standard formula without the library dependency.

Benefits of the Final Pipeline

- Reproducible and leakage-safe: every learned statistic (medians, percentile caps, scaler parameters, encoders) is fit once and reapplied consistently to new data.
- Business-interpretable: every engineered feature and drop decision is tied to an explicit business rationale, not just a statistical threshold.
- Model-agnostic: the output is usable by linear, tree-based, and distance-based model families alike.

Assumptions & Limitations

- Missingness was assumed MAR based on similar attrition rates between missing/non-missing subgroups; a formal missingness-mechanism model could further validate this.
- Percentile-based capping bounds (1st/99th) are a reasonable default, not a domain-validated business threshold — a compensation specialist could refine these further.
- Feature relevance rankings (Mutual Information, Random Forest importance) were used purely as a selection heuristic; no predictive-accuracy claims are made anywhere in this pipeline, consistent with the assignment's evaluation scope.
- The reusable pipeline assumes future data arrives in the same raw schema; upstream schema changes would require updating the ColumnTransformer's column lists.